
From Noise to Diversity: Random Embedding Injection in LLM Reasoning

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Abstract

Recent *soft prompt* methods, which prepend or insert trainable continuous embeddings into LLM inputs as virtual tokens, optimize these embeddings to enhance mathematical reasoning. However, the mechanism by which embedding injection itself affects model behavior, independent of optimization, remains unexplored. We investigate Random Soft Prompts (RSPs), continuous vectors sampled from the model’s token embedding distribution and injected without any training. Empirically, RSPs increase early-stage token entropy while generation stabilizes thereafter, and combining RSPs with temperature sampling yields higher trajectory diversity as measured by Pass@ n . Theoretically, we provide two complementary analyses: an energy-based framework showing that RSP introduces Langevin-like perturbations on the Hopfield energy landscape with natural dual decay across layers and generation steps, and a distributional analysis showing that RSP induces stochastic token rankings and expands the output support beyond temperature sampling. We further discuss the implications for GRPO training, where trajectory diversity is critical for learning signal quality.

1 Introduction

As large language models (LLMs) scale to billions of parameters, full fine-tuning becomes increasingly expensive. This has motivated parameter-efficient methods that adapt frozen models by introducing a small number of trainable parameters [Hu et al., 2022, Housby et al., 2019]. Among these, *soft prompts* prepend or insert learnable continuous embeddings into the input sequence, steering model behavior through the attention mechanism [Li and Liang, 2021, Lester et al., 2021]. This paradigm has been actively adopted for enhancing mathematical reasoning in LLMs [Kang et al., 2026, Xu et al., 2025, Ye et al., 2026, Hao et al., 2025, Zhang et al., 2026]. Despite their diverse designs, these methods share a common structure: they inject continuous embeddings that lie outside the pretrained vocabulary and optimize them to improve downstream performance.

However, we find that optimization is not a necessary condition for these effects. We observe that injecting randomly initialized, untrained continuous embeddings can also shift the output distribution. For instance, applying a chat template to a base model not fine-tuned with such format causes a prompt-format mismatch that degrades reasoning capability, but appending random embeddings to such mismatched inputs mitigates this degradation. This suggests that injecting out-of-vocabulary continuous embeddings itself, independent of any learned content, plays a non-trivial role in shaping model behavior. Existing studies primarily evaluate *soft prompt* methods through end-task accuracy, and the mechanisms by which embedding injection influences the output distribution remain insufficiently explored.

In this work, we investigate the role of Random Soft Prompts (RSPs), continuous vectors sampled from the Gaussian distribution of the model’s token embedding matrix and injected without any

training. Empirically, RSPs increase early-stage token entropy while the output stabilizes in later generation stages, and combining RSPs with temperature sampling yields more diverse reasoning trajectories as measured by Pass@ n scaling. Theoretically, we analyze RSPs from two perspectives: an energy-based framework showing that RSP introduces controlled stochastic perturbations with natural decay through the attention mechanism, and a distributional analysis showing that RSP induces stochastic token rankings and expands the output support beyond what temperature sampling can achieve. We further discuss the implications of these findings for Group Relative Policy Optimization (GRPO) [Shao et al., 2024], where trajectory diversity is essential for effective learning signals.

Our contributions are as follows:

- We empirically analyze how RSPs affect LLM generation, showing that RSPs increase early token entropy and produce more diverse reasoning trajectories when combined with temperature sampling.
- We provide theoretical analyses from two perspectives: an energy-based framework explaining why RSP can perturb the output distribution without degrading performance, and a distributional analysis showing that RSP induces stochastic token rankings and expands the output support beyond the reach of temperature sampling.
- We discuss the implications of RSP-induced diversity for GRPO training and argue that RSP provides exploration gains complementary to existing temperature-based approaches.

2 Background

2.1 Soft Prompt Methods for LLM Reasoning

Soft prompts emerged as a parameter-efficient alternative to full fine-tuning. Prefix-Tuning [Li and Liang, 2021] introduced the paradigm by prepending trainable continuous vectors to every transformer layer, Prompt Tuning [Lester et al., 2021] simplified this to input-layer-only tuning and showed that it matches full fine-tuning at sufficient scale, and P-tuning [Liu et al., 2024] extended the approach with an LSTM-based prompt encoder to model inter-position dependencies.

More recently, *soft prompts* have been actively applied to LLM reasoning. TTSV [Kang et al., 2026] optimizes prefix embeddings at test time via entropy minimization, steering the model toward high-certainty states to activate latent reasoning capabilities. SoftCoT [Xu et al., 2025] replaces explicit chain-of-thought tokens with soft tokens generated by an assistant model, leveraging the stronger representational capacity of continuous representations over discrete language. LTPO [Ye et al., 2026] optimizes latent thought embeddings per test instance through policy gradient with a confidence-based intrinsic reward. COCONUT [Hao et al., 2025] feeds the model’s own hidden states back as input embeddings, enabling reasoning in a continuous latent space. MemGen [Zhang et al., 2026] demonstrates that embedding vectors can serve as experience memory by constructing latent token sequences that enrich the model’s reasoning. Pause tokens [Goyal et al., 2024] insert learnable tokens into the input to provide additional computation steps.

These approaches all inject out-of-vocabulary continuous embeddings optimized for task-specific objectives. Because evaluation centers on end-task accuracy, the contribution of the injection itself and that of the optimization are not separated, and the mechanisms by which such out-of-vocabulary continuous embeddings shape model behavior remain a separate and underexplored question.

2.2 Noise Injection in Neural Networks

Neural networks have incorporated noise at various stages. During training, dropout [Srivastava et al., 2014] randomly deactivates hidden units to prevent overfitting, and NEFTune [Jain et al., 2024] adds noise to token embeddings during instruction fine-tuning, reporting improvements in instruction following. These methods use noise as a regularizer and do not apply perturbations at inference time.

At inference time, perturbation-based approaches primarily target robustness and safety. Randomized smoothing [Cohen et al., 2019] injects Gaussian noise into inputs to obtain certified robustness guarantees. In the LLM setting, SmoothLLM [Robey et al., 2025] perturbs input prompts at the character level to defend against adversarial jailbreaking. These methods require multiple noisy forward passes with output aggregation, and target prediction stability rather than output diversity.

Table 1: Accuracy (%) across model configurations and benchmarks. RSP reports the best result across injection positions (Prefix, Infix, Suffix). Full per-position breakdown is in Appendix A. Values in parentheses indicate the difference from the baseline.

Model	MATH-500		GSM8K		AIME24	
	Baseline	RSP	Baseline	RSP	Baseline	RSP
<i>Instruct Models</i>						
LLaMA-3.1-8B-Inst	52.20	49.40 (−2.8)	85.75	86.73 (+1.0)	6.67	10.00 (+3.3)
Qwen2.5-Math-7B-Inst	83.20	84.20 (+1.0)	95.45	95.68 (+0.2)	13.33	16.67 (+3.3)
Qwen2.5-Math-1.5B-Inst	73.00	75.20 (+2.2)	85.29	85.75 (+0.5)	10.00	20.00 (+10.0)
<i>Base Models (Raw Text)</i>						
Qwen2.5-Math-7B	72.20	71.60 (−0.6)	84.15	84.31 (+0.2)	6.67	16.67 (+10.0)
Qwen2.5-Math-1.5B	61.40	64.40 (+3.0)	77.86	74.30 (−3.6)	16.67	10.00 (−6.7)
<i>Base Models + ChatML (Format Mismatch)</i>						
Qwen2.5-Math-7B	52.20	70.40 (+18.2)	58.30	75.97 (+17.7)	23.33	23.33 (+0.0)
Qwen2.5-Math-1.5B	34.40	63.40 (+29.0)	37.45	67.02 (+29.6)	13.33	20.00 (+6.7)

In reinforcement learning, NoisyNet [Fortunato et al., 2018] injects learnable noise into network weights to improve exploration, demonstrating that the location and structure of noise injection affect exploration quality. This motivation is closest to RSPs, which similarly target diversity, but differ in injection target (input embeddings vs. network weights) and require no learned noise parameters.

RSPs differ from these approaches in the following respects. **(1)** RSPs operate solely at inference without any training, whereas dropout and NEFTune inject noise during training, modifying the resulting model parameters accordingly. **(2)** RSPs preserve the original input and append fresh embedding vectors in a single forward pass, whereas robustness methods corrupt the input and aggregate over multiple noisy passes. **(3)** RSPs use purely random vectors sampled from the pretrained embedding statistics, whereas NoisyNet learns noise parameters through optimization.

3 Experimental Analysis

3.1 Random Soft Prompts: Definition and Experimental Setup

Let $\mathbf{W}_E \in \mathbb{R}^{V \times d}$ denote the token embedding matrix of a pretrained LLM, where V is the vocabulary size and d is the hidden dimension. We define an RSP $\mathbf{H} = [h_1, \dots, h_L] \in \mathbb{R}^{L \times d}$ as a sequence of L continuous vectors, where each vector is independently sampled from an isotropic Gaussian distribution whose mean and variance are matched to those of the pretrained token embedding matrix:

$$h_j \sim \mathcal{N}(\mu_E, \sigma_E^2 \mathbf{I}), \quad j = 1, \dots, L, \quad (1)$$

with $\mu_E = \frac{1}{Vd} \sum_{i,k} [\mathbf{W}_E]_{i,k}$ and $\sigma_E = \text{std}(\mathbf{W}_E)$. Let $\mathbf{X} = [x_1, \dots, x_T] \in \mathbb{R}^{T \times d}$ denote the embedding sequence of the original input. The RSP-augmented input $\tilde{\mathbf{X}}$ is constructed by concatenating $\mathbf{H}$ with $\mathbf{X}$ at one of three positions without modifying the original prompt text: *prefix* ($[\mathbf{H}; \mathbf{X}]$), *infix* ($\mathbf{H}$ inserted within $\mathbf{X}$), or *suffix* ($[\mathbf{X}; \mathbf{H}]$), following the injection strategies adopted in prior work [Kang et al., 2026, Xu et al., 2025, Ye et al., 2026, Zhang et al., 2026]. Sampling a new $\mathbf{H}$ for each batch ensures that the results are not dependent on any realization of the random embeddings.

We evaluate on mathematical reasoning benchmarks: MATH-500 [Lightman et al., 2024], GSM8K [Cobbe et al., 2021], and AIME24, using LLaMA-3.1-8B-Instruct [Grattafiori et al., 2024] and the Qwen2.5-Math model family [Yang et al., 2024] across instruct models, base models, and base models with mismatched chat templates. All experiments use greedy decoding to isolate the effect of RSPs from sampling stochasticity. Full experimental details are provided in Appendix A.

3.2 How Does Untrained RSP Compare to Baselines & Optimized Soft Prompts?

We first examine how injecting random embeddings affects the model’s reasoning performance. Table 1 reports the accuracy for each model configuration, using the best RSP result across injection positions (Prefix, Infix, Suffix) and $L \in \{10, 15, 20\}$ for each benchmark.

Table 2: Comparison with existing soft prompt methods (TTSV, SoftCoT, LTPO) under unified evaluation. Baseline and RSP values are from Table 1. **Bold** denotes the best result and underline denotes the second best for each model–benchmark pair.

Model	Benchmark	Baseline	RSP	TTSV	SoftCoT	LTPO
Qwen2.5-Math-7B-Instruct	MATH-500	83.20	84.20	<u>83.80</u>	82.80	83.00
	GSM8K	<u>95.45</u>	95.68	95.30	95.00	94.84
	AIME24	13.33	<u>16.67</u>	23.30	<u>16.67</u>	13.33
Qwen2.5-Math-1.5B-Instruct	MATH-500	73.00	<u>75.20</u>	<u>75.20</u>	76.00	72.40
	GSM8K	85.29	85.75	<u>85.70</u>	84.46	84.53
	AIME24	<u>10.00</u>	20.00	6.70	6.67	<u>10.00</u>

For instruct models and base models with their native prompt formats, RSP injection maintains or slightly shifts baseline performance within a few percentage points. Most strikingly, under prompt-format mismatch — where base models receive ChatML templates they were not trained on — RSP yields pronounced improvements of up to +29 pp. These gains are difficult to attribute to simple perturbation effects, as noise that merely raises model uncertainty would further degrade performance.

We compare RSPs with prior soft prompt methods (TTSV, SoftCoT, LTPO) that optimize their embeddings for distinct objectives, under a unified evaluation protocol reported in Table 2. All methods are re-evaluated with the same unified extraction pipeline (Appendix A.2), enabling a format-agnostic comparison despite each method’s distinct prompt format and grading scheme. The results place RSPs on par with the optimized methods without requiring any training.

3.3 How Does RSP Affect the Output Distribution?

We analyze how RSP injection changes the model’s output distribution on Qwen2.5-Math-1.5B-Instruct with MATH-500, measuring per-token entropy, top-1 probability, and varentropy across the generation process. Figure 1 compares these metrics for the first 5% of generation steps across RSP variants and prior soft prompt methods (LTPO, SoftCoT, TTSV). Full statistics are in Appendix A.4.

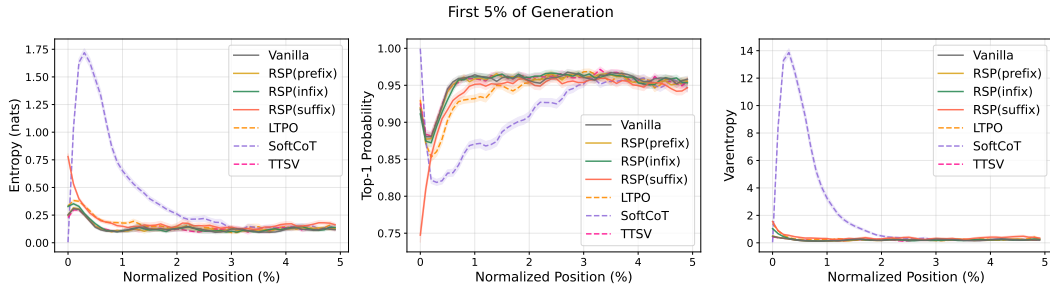


Figure 1: Mean entropy, top-1 probability, and varentropy during the first 5% of generation steps (Qwen2.5-Math-1.5B-Instruct, MATH-500). Shading indicates ± 1 SEM. Solid lines: RSP variants and the baseline. Dashed lines: prior soft prompt methods (LTPO, SoftCoT, TTSV).

These results show that RSP modifies the early-stage output distribution across all injection positions: entropy increases (the token distribution flattens), top-1 probability decreases (commitment to a dominant token weakens), and varentropy increases (indicating multimodal distributions [Ahmed et al., 2026]). High-entropy positions are known to act as critical forking points that determine reasoning trajectories [Wang et al., 2025], so this shift moves the model toward a state where multiple reasoning paths can coexist. The change is localized to the initial reasoning stage. In the middle and final portions of generation, all three metrics converge to baseline levels.

Although existing soft prompt methods are optimized for different objectives, they share a common pattern of increasing early-stage entropy. Notably, TTSV, which optimizes soft prompts using a reward that reduces the mean entropy of reasoning trajectories, still exhibits early entropy comparable to the baseline. This suggests that elevating early-stage entropy is an inherent effect of injecting non-pretrained continuous embeddings, independent of the optimization objective.

3.4 Does RSP Induce Trajectory Diversity?

Section 3.3 empirically established that RSP reshapes the early-stage output distribution. We now examine whether this shift translates into reasoning diversity at three complementary levels of analysis: token rankings, semantic content of trajectories, and task-level outcomes. The three analyses converge on a directional picture: RSP’s first-token perturbation propagates into semantically more diverse reasoning trajectories, which in turn yields greater output diversity at the task level. All three analyses share the setting: Qwen2.5-Math-1.5B-Instruct, MATH-500, suffix injection, and $L = 10$.

Token-level: distribution beyond temperature. We test whether RSP modifies the first-token distribution in a qualitatively different way from temperature scaling, which preserves token rankings while only flattening probabilities. Under autoregressive decoding, any trajectory-level divergence between RSP and the baseline must originate from the first generation step, a critical forking point for reasoning trajectories [Wang et al., 2025]. Using the baseline at $\tau=1.0$ as reference, we compare RSP at $\tau=1.0$ against the baseline at $\tau=2.0$

(the high-temperature ceiling of what flattening alone can achieve) on three complementary metrics: Spearman rank correlation ρ to detect rank reorganization, the probability mass placed outside the baseline’s top-10 to measure support expansion, and Jensen–Shannon divergence to quantify the overall magnitude of distributional change (formal definitions in Appendix B).

Table 3 reports the contrast. Even at $\tau=2.0$, only 5.15% of probability mass falls outside the baseline’s top-10. RSP at $\tau=1.0$ reaches 27.64%. The Spearman rank correlation tells the story: because temperature scaling $p_i^{(\tau)} \propto p_i^{1/\tau}$ is a monotone transform that preserves token ranking ($p_i > p_j \iff p_i^{(\tau)} > p_j^{(\tau)}$) for any $\tau > 0$, ρ relative to the baseline is exactly 1. RSP, by contrast, reranks the token distribution itself, reducing ρ to 0.651—an effect temperature cannot produce at any τ . The Jensen–Shannon divergence (0.231) quantifies the overall magnitude of this distributional change. Since $\rho < 1$ rules out rank-preserving flattening as the sole cause, the high divergence reflects a structural reshape of the distribution rather than a uniform flattening of the same ranking.

Table 3: First-token distribution metrics measured against the baseline at $\tau=1.0$, comparing the baseline at $\tau=2.0$ with RSP at $\tau=1.0$.

Metric	$\tau=2.0$	RSP
Spearman ρ	1.0	0.651
Mass outside top-10	5.15%	27.64%
JS divergence	0.086	0.231

Trajectory-level: semantic diversity. Token-level reranking raises a second question: do first-token perturbations produce semantically diverse reasoning trajectories, or do independently perturbed trajectories converge to semantically similar ones? We randomly sample 64 problems from MATH-500 and generate 300 independent trajectories per problem under Baseline and RSP at temperature $\tau = 1.0$. Each trajectory is encoded into a dense semantic vector by BGE-M3 [Chen et al., 2024], a sentence-level multilingual embedding model, and we compute three complementary metrics over these embeddings: *pairwise cosine distance* between trajectories captures the overall semantic spread of the trajectory set; *inter-cluster distance* between centroids identified by DBSCAN [Ester et al., 1996], a density-based clustering algorithm that groups nearby trajectories without prespecifying the number of clusters, captures the separation between distinct trajectory groups; and *intra-cluster distance* captures semantic coherence within each group.

Table 4 reveals three concurrent patterns. Pairwise cosine distance rises by about $1.4\times$, indicating that the trajectory set as a whole becomes more semantically diverse. Inter-cluster distance jumps by about $5.4\times$, showing that the trajectories split into substantially more separated groups. Intra-cluster distance is essentially unchanged, indicating that semantic coherence within each group remains comparable to the baseline. RSP also yields larger pairwise distance than baseline on 56 of 64 problems (87.5%).

Table 4: Semantic diversity metrics (64 problems, 300 trajectories per condition).

Metric	Baseline	RSP
Pairwise cos. dist.	0.0612	0.0879
Inter-cluster dist.	0.0183	0.0982
Intra-cluster dist.	0.0526	0.0528

Outcome-level: Pass@ n scaling. Finally, we evaluate whether the token- and trajectory-level diversity translates into task-level outcome diversity through Pass@ n [Chen et al., 2021], the probability that at least one of n sampled solutions is correct. We compare four conditions with $n = 16$ samples per problem on MATH-500 and AIME24: (i) *Baseline*, $\tau=1.0$: temperature sampling only;

198 (ii) *RSP (fixed seed)*, $\tau = 1.0$: a single RSP shared across all samples; (iii) *RSP (indep. seed)*, greedy:
 199 a different RSP per sample without temperature; and (iv) *RSP (indep. seed)*, $\tau = 1.0$: a different RSP
 200 per sample combined with temperature sampling.

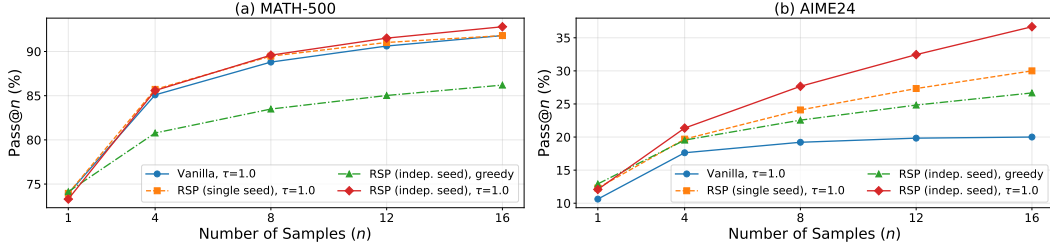


Figure 2: Pass@ n scaling on (a) MATH-500 and (b) AIME24 with Qwen2.5-Math-1.5B-Instruct, $n = 16$ samples per problem. Baseline: temperature sampling only. Fixed seed: single RSP shared across samples combined with temperature. Indep. seed: a different RSP per sample, with or without temperature.

201 Condition (iv) achieves the highest Pass@ n on both benchmarks, reaching 92.80% on MATH-500 and
 202 36.67% on AIME24 at $n = 16$. Condition (ii) shows no improvement over the baseline, suggesting
 203 that the diversity gain likely depends on varying the random embeddings across samples rather
 204 than a single fixed perturbation. Pass@1 remains comparable to the baseline on MATH-500, and
 205 on the more challenging AIME24, (iv) achieves higher Pass@1 than the baseline. Together these
 206 observations suggest that combining independent RSPs with temperature sampling may yield greater
 207 trajectory diversity while preserving single-sample accuracy.

208 4 Theoretical Analysis: Why RSP Works

209 4.1 RSP as Controlled Exploration on the Energy Landscape

210 Under the two simplifying assumptions introduced below, this section provides a theoretical account
 211 of two properties observed in our experiments: (1) RSP can change the model’s output distribution,
 212 and (2) RSP does not degrade reasoning performance. We show that RSP converts the transformer’s
 213 deterministic forward pass into a controlled stochastic process on the Hopfield energy landscape,
 214 where the noise introduced through attention naturally decays with both layer depth and generation
 215 step. We build on the equivalence between softmax attention and gradient descent on the Modern
 216 Hopfield energy [Ramsauer et al., 2021]. The analysis relies on two simplifying assumptions:
 217 (A1) $\mathbf{K} = \mathbf{V}$ in attention (exact in Hopfield models, approximate in standard transformers) and
 218 (A2) weight-tied layers (exact in deep equilibrium models [Bai et al., 2019], approximate otherwise).
 219 Full proofs are in Appendix C.

220 **Transformer layers as energy minimization.** The Modern Hopfield Energy for stored patterns
 221 $\Xi \in \mathbb{R}^{d \times m}$ and state $\sigma \in \mathbb{R}^d$ is $E(\sigma; \Xi) = -\beta^{-1} \log \sum_{j=1}^m \exp(\beta \xi_j^\top \sigma) + \frac{1}{2} \|\sigma\|^2$, where $\beta =$
 222 $1/\sqrt{d}$. A full gradient step on this energy yields $\sigma^{\text{new}} = \Xi \cdot \text{softmax}(\beta \Xi^\top \sigma)$, which is identical to
 223 softmax attention [Ramsauer et al., 2021]. Each transformer layer thus performs one gradient descent
 224 step on this energy landscape: the hidden state moves toward a lower-energy configuration at every
 225 layer.

226 **The trapping problem.** This monotone descent implies that, without perturbation, the forward
 227 pass converges to the local minimum in the basin of attraction of $\sigma^{(0)}$. If a better minimum σ_g^* exists
 228 in a different basin, the deterministic forward pass cannot reach it because it cannot cross energy
 229 barriers.

230 **Why temperature scaling and additive noise do not resolve this.** Temperature scaling modi-
 231 fies only the final output distribution: $p^{(\tau)}(v) = \text{softmax}(\mathbf{z}^{(L)}/\tau)$, leaving the internal trajectory
 232 $\mathbf{x}^{(l+1)}|_\tau = \mathbf{x}^{(l+1)}|_{\tau=1}$ for all $l < L$ identical. It cannot help escape a local minimum during inference.
 233 Additive Gaussian noise $\mathbf{x}^{(0)} + \epsilon$ perturbs the initial state, but in subsequent layers the perturbation

234 magnitude is bounded by $\prod_{l'} L_{f_{l'}} \cdot \sigma$ where $L_{f_{l'}}$ is the Lipschitz constant of layer l' . When $L_{f_{l'}} > 1$,
 235 the noise amplifies rather than decays, providing no controlled exploration schedule.

236 **RSP as controlled exploration via attention.** RSP resolves both limitations through two mecha-
 237 nisms.

238 First, concatenating k random vectors sampled from $\mathcal{N}(\mu_E, \sigma_E^2 \mathbf{I})$ causes each token to attend to both
 239 real and random keys, yielding

$$\mathbf{x}_i^{(l+1)} = (1 - w_{r,i}^{(l)}) \tilde{\mathbf{x}}_i^{(l+1)} + \boldsymbol{\eta}_i^{(l)}, \quad (2)$$

240 where $\tilde{\mathbf{x}}_i^{(l+1)}$ is the standard attention update over real tokens, and $\boldsymbol{\eta}_i^{(l)} := \sum_{j=1}^k \alpha_{i,n+j}^{(l)} \mathbf{v}_{r_j}^{(l)}$ aggre-
 241 gates the contribution of random tokens with total attention mass $w_{r,i}^{(l)} := \sum_{j=1}^k \alpha_{i,n+j}^{(l)}$.

242 The perturbation magnitude is controlled by $w_{r,i}^{(l)}$, with variance scaling on the order of $(w_{r,i}^{(l)})^2$. In
 243 practice, since the embedding mean is negligible relative to its standard deviation ($|\mu_E|/\sigma_E \ll 1$),
 244 the bias term is dominated by the stochastic component, and the dynamics are well-approximated by
 245 a gradient descent process with approximately zero-mean structured noise (see Appendix C.2).

246 This yields a Langevin-like stochastic process that enables exploration beyond local minima.

247 Second, RSP modifies the energy landscape. Adding k random keys changes the Hopfield energy as

$$\tilde{E}_i - E_i = -\beta^{-1} \log \left(1 + \frac{B_i}{A_i} \right) \leq 0, \quad (3)$$

248 where $A_i = \sum_j \exp(\beta \mathbf{k}_j^\top \mathbf{q}_i)$ and $B_i = \sum_j \exp(\beta \mathbf{k}_{r_j}^\top \mathbf{q}_i) > 0$. The magnitude scales with
 249 $w_{r,i}^{(l)}/(1 - w_{r,i}^{(l)})$, so tokens assigning more attention to random keys experience larger energy pertur-
 250 bations.

251 While this does not directly reduce barrier height, it alters the local energy landscape, and together
 252 with stochastic perturbations increases the probability of crossing energy barriers between basins.

253 Both mechanisms are governed by $w_{r,i}^{(l)}$, which controls the overall strength of exploration.

254 **Why performance is preserved: dual decay of $w_{r,i}^{(l)}$.** The attention decay bound is:

$$w_{r,i}^{(l)} \leq \frac{k}{k + n \cdot \exp(\Delta_i^{(l)}/\sqrt{d})}, \quad (4)$$

255 where $\Delta_i^{(l)} := \min_{j \in [n]} \mathbf{q}_i^\top \mathbf{k}_j - \max_{j' \in [k]} \mathbf{q}_i^\top \mathbf{k}_{r_{j'}}$ is the real-vs-random logit gap and n is the
 256 number of real tokens. Since the bias, stochastic scale, and energy modification are all controlled by
 257 $w_{r,i}^{(l)}$, this bound governs the overall perturbation strength.

258 **Layer-wise decay.** If deeper layers increase the real-vs-random logit gap $\Delta_i^{(l)}$, then $w_{r,i}^{(l)}$ correspond-
 259 ingly decays toward zero. Empirically, as deeper layers develop task-specific representations, $\Delta_i^{(l)}$
 260 tends to increase, making RSP’s perturbation strongest in early layers and negligible in deeper layers.

261 **Sequence-wise decay.** In autoregressive generation, each newly generated token adds a real key to
 262 the KV cache, so n increases at every generation step while k remains fixed. The growing n in Eq. (4)
 263 causes $w_{r,i}^{(l)}$ to decrease monotonically, progressively diluting RSP’s influence as the sequence grows.

264 This dual decay provides a natural annealing schedule: RSP enables exploration in the early stage
 265 of both the internal forward pass and the generation sequence, while the direct influence of the
 266 perturbation progressively diminishes as the model accumulates real tokens and converges to task-
 267 specific representations.

268 **Empirical verification.** We empirically verify this dual decay on Qwen2.5-Math-7B with suffix
 269 injection over 500 MATH-500 problems. For each problem, we measure the per-token attention
 270 mass directed to the RSP embeddings and to the question tokens within the same sequence, with the

question tokens serving as a baseline that factors out sequence-length effects. Figure 3 shows the expected contrast: the question tokens, which remain task-relevant throughout reasoning, receive attention whose distribution varies across layers and reasoning positions without a consistent trend, while the RSP embeddings, which carry no task-relevant content and act as noise, receive attention that decreases along both the layer and sequence axes, directly instantiating the layer-wise and sequence-wise dilution predicted by Eq. (4).

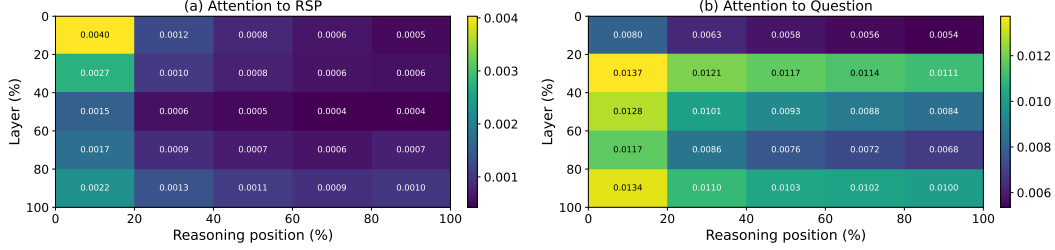


Figure 3: Per-token attention mass on Qwen2.5-Math-7B (suffix, 500 MATH-500 problems, each with an independently sampled RSP). The layer and reasoning-position axes are each partitioned into five quantile bins. Every cell reports the 500-sample mean per-token attention mass directed to (a) the RSP embeddings and (b) the question tokens within that region. The horizontal axis spans reasoning position from early (left) to late (right) generation steps, and the vertical axis spans layer depth from shallow (top) to deep (bottom). Attention to the RSP embeddings in (a) decays along both axes — concentrating at shallow layers and early reasoning positions — while attention to the question tokens in (b) remains comparatively uniform, visually confirming the layer-wise and sequence-wise dilution predicted by Eq. (4).

4.2 Output Distribution Shift and Exploration

We study how Random Soft Prompting (RSP) alters the output distribution of an autoregressive language model and how this leads to improved exploration.

In autoregressive decoding, generation proceeds via prefix-conditioned distributions $p(y_t | x, y_{<t})$. RSP does not modify a single global distribution, but perturbs the sequence of conditional distributions across decoding steps. In Section 4.1, we analyzed this from an internal, layer-wise and sequence-wise perspective, showing that RSP introduces structured perturbations through attention that naturally decay across layers. We now examine how these perturbations manifest at the output level.

Limitation of temperature scaling. Temperature scaling transforms probabilities as $p_i^{(\tau)} \propto p_i^{1/\tau}$ and preserves token ranking, i.e., $p_i > p_j \iff p_i^{(\tau)} > p_j^{(\tau)}$. Hence it cannot introduce new token orderings at a fixed decoding state.

Effect of RSP. RSP induces stochastic perturbations to the hidden state, which we locally approximate as $z_i(h) \approx z_i + a_i^\top h$, where h is a random soft prompt.

Proposition 1 (Stochastic ranking). If there exists a token pair (i, j) such that $a_i \neq a_j$, then

$$0 < \Pr_h(z_i(h) > z_j(h)) < 1.$$

Thus, unlike temperature scaling, RSP randomizes token rankings at each decoding step.

Proposition 2 (Support expansion). Let $\text{TopK}_{\text{Temp}}(s_t)$ be the top- K tokens under temperature scaling at state s_t . Under the same condition, there exists $i \notin \text{TopK}_{\text{Temp}}(s_t)$ such that

$$\Pr_h(i \in \text{TopK}(p^{\text{RSP}}(\cdot | s_t, h))) > 0.$$

Implication. RSP expands the effective support of the output distribution, enabling tokens that are effectively inaccessible under temperature scaling. Since autoregressive generation composes these distributions over time, such perturbations can lead to distinct trajectories, explaining the empirical gains in exploration.

5 Conclusion

Summary. This work investigated the role of Random Soft Prompts (RSPs), continuous embeddings sampled from the pretrained token embedding distribution and injected into the model input without any training. Through empirical and theoretical analyses, we established three findings. First, RSPs increase early-stage token entropy, top-1 probability reduction, and varentropy while the model’s output stabilizes in later generation stages (Section 3). Second, RSPs induce stochastic token rankings and expand the effective support of the output distribution, a property that temperature sampling fundamentally lacks due to its ranking-preserving nature (Section 4). Third, combining RSPs with temperature sampling produces more diverse reasoning trajectories than either method alone, as measured by Pass@ n scaling (Section 3).

Implications for GRPO training. GRPO computes advantages by comparing multiple sampled trajectories within each group, making trajectory diversity essential for effective learning signals. Existing approaches such as DAPO [Yu et al., 2025] and CE-GPPO [Su et al., 2026] address entropy collapse by modifying the clipping range or reintroducing gradients from clipped tokens, but the underlying sampling process remains based on temperature sampling, which preserves token rankings and cannot expand the output support. Our theoretical and empirical results suggest that RSP can address this limitation at the sampling level: by perturbing the initial output distribution with independent random embeddings per rollout, RSP provides a complementary source of diversity that is orthogonal to temperature-based exploration. We expect this to yield more effective learning signals for GRPO training, particularly in mitigating entropy collapse.

Limitations. Our experiments are conducted on mathematical reasoning benchmarks (MATH-500, GSM8K, AIME24) with a limited set of model families (Qwen2.5-Math, LLaMA-3.1). The energy-theoretic analysis in Section 4 relies on strong assumptions: the Hopfield–Attention equivalence (A1: $\mathbf{K} = \mathbf{V}$) and weight-tied layers (A2), which hold exactly only in Hopfield-type models and deep equilibrium models, respectively. Since real transformer architectures vary widely in their attention mechanisms and layer structures, the theoretical predictions are most directly applicable to models whose attention dynamics closely approximate the Hopfield energy framework. The GRPO application remains a theoretical expectation and has not been empirically validated.

Future work. Extending the experimental evaluation to other domains (e.g., code generation, commonsense reasoning) and larger model scales would test the generality of RSP’s effects. Empirically validating the GRPO application by training with RSP-augmented rollouts and comparing against DAPO and CE-GPPO is a natural next step. Investigating the optimal RSP distribution and token count, which currently require per-model tuning, could lead to more principled injection strategies.

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511 A Experimental Setup and Full Accuracy Breakdown

512 Table 5 provides the full per-position accuracy breakdown (Prefix, Infix, Suffix) corresponding to
513 the best-across-positions results summarized in Table 1 in the main text. Table 6 lists the number of
514 RSP tokens L used for each model–benchmark pair, selected from $\{10, 15, 20\}$. All experiments are
515 conducted on a single NVIDIA A6000 40GB GPU with greedy decoding, a maximum generation
516 length of 3,072 tokens for MATH-500 and GSM8K, and 4,096 tokens for AIME24. Batch size is
517 fixed per model (16 for LLaMA-3.1-8B-Instruct and Qwen2.5-Math-7B variants, 32 for Qwen2.5-
518 Math-1.5B variants).

Table 5: Accuracy (%) across model configurations, injection positions, and benchmarks. Values in parentheses indicate the difference from the baseline. **Bold** denotes the best result and underline denotes the second best for each model–benchmark pair.

Model	Mode	MATH-500	GSM8K	AIME24
<i>Instruct Models</i>				
LLaMA-3.1-8B-Instruct	Baseline	52.20	85.75	<u>6.67</u>
	Prefix	45.00 (−7.2)	76.35 (−9.4)	3.33 (−3.3)
	Infix	49.40 (−2.8)	85.97 (+0.2)	10.00 (+3.3)
	Suffix	<u>49.40</u> (−2.8)	86.73 (+1.0)	10.00 (+3.3)
Qwen2.5-Math-7B-Instruct	Baseline	83.20	95.45	<u>13.33</u>
	Prefix	81.40 (−1.8)	94.92 (−0.5)	13.33 (+0.0)
	Infix	84.20 (+1.0)	95.60 (+0.2)	16.67 (+3.3)
	Suffix	<u>83.40</u> (+0.2)	95.68 (+0.2)	16.67 (+3.3)
Qwen2.5-Math-1.5B-Instruct	Baseline	73.00	<u>85.29</u>	10.00
	Prefix	74.80 (+1.8)	85.29 (+0.0)	<u>13.33</u> (+3.3)
	Infix	75.20 (+2.2)	85.75 (+0.5)	6.67 (−3.3)
	Suffix	74.20 (+1.2)	84.69 (−0.6)	20.00 (+10.0)
<i>Base Models (Raw Text)</i>				
Qwen2.5-Math-7B	Baseline	72.20	84.15	6.67
	Prefix	<u>71.60</u> (−0.6)	84.31 (+0.2)	16.67 (+10.0)
	Infix	63.20 (−9.0)	64.29 (−19.9)	16.67 (+10.0)
	Suffix	55.60 (−16.6)	54.13 (−30.0)	<u>13.33</u> (+6.7)
Qwen2.5-Math-1.5B	Baseline	<u>61.40</u>	77.86	16.67
	Prefix	64.40 (+3.0)	74.30 (−3.6)	10.00 (−6.7)
	Infix	63.20 (+1.8)	73.92 (−3.9)	<u>10.00</u> (−6.7)
	Suffix	54.60 (−6.8)	58.61 (−19.3)	3.33 (−13.3)
<i>Base Models + ChatML (Format Mismatch)</i>				
Qwen2.5-Math-7B	Baseline	52.20	58.30	23.33
	Prefix	59.20 (+7.0)	56.41 (−1.9)	3.33 (−20.0)
	Infix	62.00 (+9.8)	69.07 (+10.8)	23.33 (+0.0)
	Suffix	70.40 (+18.2)	75.97 (+17.7)	23.33 (+0.0)
Qwen2.5-Math-1.5B	Baseline	34.40	37.45	<u>13.33</u>
	Prefix	63.40 (+29.0)	67.02 (+29.6)	20.00 (+6.7)
	Infix	39.20 (+4.8)	50.42 (+13.0)	6.67 (−6.7)
	Suffix	<u>51.00</u> (+16.6)	<u>50.64</u> (+13.2)	<u>13.33</u> (+0.0)

Table 6: Number of RSP tokens (L) used for each model and benchmark.

Model	MATH-500	GSM8K	AIME24
LLaMA-3.1-8B-Instruct	15	20	15
Qwen2.5-Math-7B-Instruct	20	20	20
Qwen2.5-Math-1.5B-Instruct	20	10	20
Qwen2.5-Math-7B	10	10	20
Qwen2.5-Math-1.5B	10	10	20
Qwen2.5-Math-1.5B (ChatML)	20	20	10
Qwen2.5-Math-7B (ChatML)	20	20	20

519 A.1 RSP Injection Positions and Prompt Templates

520 Below we show the prompt structure for each injection position across all model types. Blue text
521 indicates RSP embeddings, and purple text indicates special tokens.

522 A.1.1 Prefix

523 RSP embeddings are concatenated before the prompt embeddings. No text modification is applied.

524 ChatML (Qwen Instruct / Base + ChatML).

```
[Random Embeddings (L tokens)]
<|im_start|>system
Please reason step by step, and put your final answer within \boxed{<|im_end|>
<|im_start|>user
{question}<|im_end|>
<|im_start|>assistant
```

526 LLaMA Chat Template.

```
[Random Embeddings (L tokens)]
<|begin_of_text|>
<|start_header_id|>system<|end_header_id|>
Cutting Knowledge Date: December 2023
Today Date: 26 Jul 2024
Please reason step by step, and put your final answer within \boxed{<|eot_id|>
<|start_header_id|>user<|end_header_id|>
{question}<|eot_id|>
<|start_header_id|>assistant<|end_header_id|>
```

528 Raw Text (Qwen Base).

```
[Random Embeddings (L tokens)]
{question}
Please reason step by step, and put your final answer within \boxed{<|eot_id|>
```

530 A.1.2 Infix

531 A description text and special tokens are inserted into the prompt. The special token positions are
532 then replaced with RSP embeddings at the embedding level.

533 ChatML (Qwen Instruct / Base + ChatML).

```
<|im_start|>system
Please reason step by step, and put your final answer within \boxed{<|im_end|>
<|im_start|>user
{question}

There are {L} special tokens that contain compressed latent reasoning information that
might be useful for your reasoning. If these tokens are useful for your case, you can
use them as reference. If these tokens are not useful for your case, you can ignore
them and focus back to solving the problem.

Here are the {L} special tokens: <special_token_1>...<special_token_L>
<|im_end|>
<|im_start|>assistant
```

535 LLaMA Chat Template.

```
<|begin_of_text|>
<|start_header_id|>system<|end_header_id|>
Cutting Knowledge Date: December 2023
Today Date: 26 Jul 2024
Please reason step by step, and put your final answer within \boxed{<|eot_id|>
<|start_header_id|>user<|end_header_id|>
```


{question}

There are $\{L\}$ special tokens that contain compressed latent reasoning information that might be useful for your reasoning. If these tokens are useful for your case, you can use them as reference. If these tokens are not useful for your case, you can ignore them and focus back to solving the problem.

Here are the $\{L\}$ special tokens:

```
<reserved_special_token_0>...  
<reserved_special_token_L>  
<|eot_id|>  
<|start_header_id|>assistant<|end_header_id|>
```

537

538 Raw Text (Qwen Base).

{question}

There are $\{L\}$ special tokens that contain compressed latent reasoning information that might be useful for your reasoning. If these tokens are useful for your case, you can use them as reference. If these tokens are not useful for your case, you can ignore them and focus back to solving the problem.

Here are the $\{L\}$ special tokens: $\langle \text{special_token_1} \rangle \dots \langle \text{special_token_L} \rangle$

Please reason step by step, and put your final answer within `\boxed{\}`.

539

540 A.1.3 Suffix

541 For chat-templated models, RSP embeddings are inserted immediately before the end-of-turn token.

542 For raw text models, RSP embeddings are concatenated at the end of the prompt.

543 ChatML (Qwen Instruct / Base + ChatML).

```
<|im_start|>system  
Please reason step by step, and put your final answer within \boxed{ }.<|im_end|>  
<|im_start|>user  
{question}[Random Embeddings ( $L$  tokens)]<|im_end|>  
<|im_start|>assistant
```

544

545 LLaMA Chat Template.

```
<|begin_of_text|>  
<|start_header_id|>system<|end_header_id|>  
Cutting Knowledge Date: December 2023  
Today Date: 26 Jul 2024  
Please reason step by step, and put your final answer within \boxed{ }.<|eot_id|>  
<|start_header_id|>user<|end_header_id|>  
{question}[Random Embeddings ( $L$  tokens)]<|eot_id|>  
<|start_header_id|>assistant<|end_header_id|>
```

546

547 Raw Text (Qwen Base).

```
{question}  
Please reason step by step, and put your final answer within \boxed{ }. [Random  
Embeddings ( $L$  tokens)]
```

548

549 A.2 Answer Extraction and Grading

550 The final answer is extracted from the model output through a fallback chain. Each step is attempted
551 in order; if extraction fails, the next step is tried.

Answer Extraction Fallback Chain

1. "final answer is \$...\$. I hope" pattern (Minerva format)
2. `\boxed{...}`: last non-empty box is used; empty `\boxed{}` are skipped
3. Text following "the answer is"
4. Text following "final answer is"
5. Last number in the output (regex fallback)

Extracted answers are normalized by removing \$, \left, \right, and unit strings. Grading is performed via symbolic equivalence checking: exact string match, numerical comparison (`math.isclose, rel_tol=1e-4`), and SymPy symbolic simplification.

A.3 Reproduced Prior Work Hyperparameters

All prior methods are reproduced on Qwen2.5-Math-1.5B-Instruct and Qwen2.5-Math-7B-Instruct, evaluated with greedy decoding (temperature = 0) and our unified answer extraction pipeline.

TTSV. Prefix length 20, trained for 20 epochs with AdamW optimizer, batch size 2, gradient accumulation 8 steps, and linear learning rate schedule. Learning rate is 1e-3 for Qwen-1.5B and 1e-5 for Qwen-7B.

LTPO. Table 7 reports the per-benchmark hyperparameters.

Table 7: LTPO hyperparameters.

Parameter	GSM8K	MATH-500	AIME24
tokens	8	8	8
steps	20	20	20
top_k	10	10	10
sigma	20	20	5
sigma_decay	0.95	0.95	0.95
lr	1e-2	5e-3	5e-2

SoftCoT. Projection module trained for 10 epochs. Thought tokens: 32 during training, 4 during evaluation. Batch size 4 for GSM8K, 1 for MATH with gradient accumulation 4. The small (assistant) model is Qwen2.5-Math-1.5B-Instruct in both configurations, while the large model is Qwen2.5-Math-1.5B-Instruct (learning rate 2e-5) or Qwen2.5-Math-7B-Instruct (learning rate 5e-6). For MATH-500 and AIME24 evaluation, we use the projection module trained on GSM8K, as it yields higher accuracy than the one trained on MATH.

A.4 Full Entropy Curves

Figure 4 shows the full normalized (0–100%) generation trajectories for entropy, top-1 probability, and varentropy, and Table 8 reports the corresponding scalar statistics including overall means, first-5% means, and the average number of generated tokens per problem. All methods converge to similar levels after the initial generation stage, confirming that the distributional changes induced by RSP are localized to the early reasoning phase.

Table 8: Per-step statistics for Qwen2.5-Math-1.5B-Instruct on MATH-500 across the full generation and the first 5% of generation steps, together with the average number of generated tokens per problem. Top-1 Prob (overall) is reported as a weighted average over the first-10%/middle-last-10% segment means with weights 0.1/0.8/0.1. Extension of Figure 1 (main text) and Figure 4.

Metric	Baseline	Prefix	Infix	Suffix	LTPO	SoftCoT	TTSV
Tokens (mean)	575.7	558.4	549.9	573.3	592.3	594.4	577.4
Entropy (first 5%)	0.1332	0.1366	0.1402	0.1941	0.1662	0.4218	0.1359
Entropy (overall)	0.0871	0.0881	0.0899	0.1049	0.0909	0.1059	0.0864
Top-1 Prob (first 5%)	0.9535	0.9523	0.9525	0.9373	0.9437	0.9156	0.9534
Top-1 Prob (overall)	0.9684	0.9681	0.9678	0.9647	0.9683	0.9651	0.9685
Varentropy (first 5%)	0.2181	0.2207	0.2463	0.4010	0.2953	2.2234	0.2174
Varentropy (overall)	0.1353	0.1375	0.1422	0.2038	0.1452	0.2404	0.1361

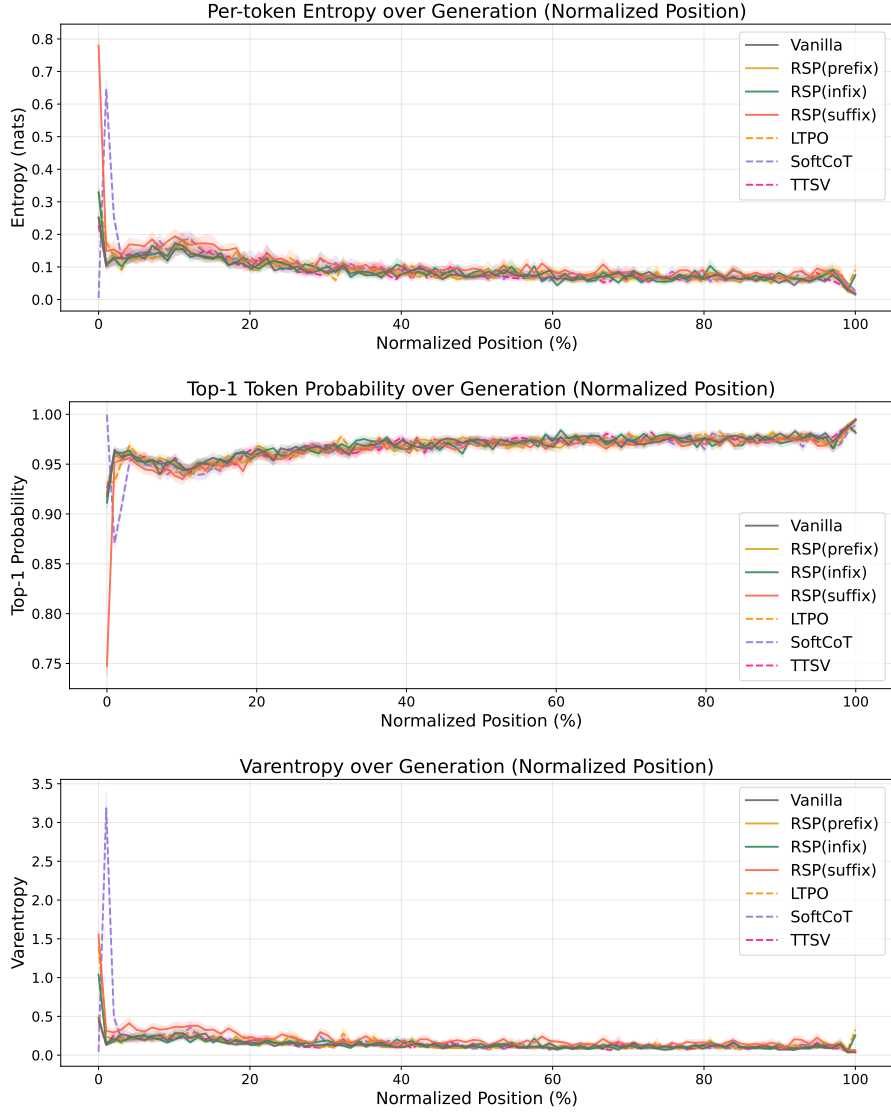


Figure 4: Mean entropy (top), top-1 probability (middle), and varentropy (bottom) over the full normalized generation trajectory (0–100%). Averaged over MATH-500, shading indicates ± 1 SEM. Solid lines: RSP variants and the baseline. Dashed lines: existing soft prompt methods. All methods converge after the initial stage.

575 B Token-Level Distribution Metrics

576 This appendix provides formal definitions for the three metrics used in the Token-level analysis of
 577 Section 3.4. Let P_{base} denote the baseline next-token distribution at $\tau = 1.0$ and P_{target} the candidate
 578 distribution being compared (e.g., the baseline at $\tau = 2.0$ or RSP at $\tau = 1.0$). All metrics are
 579 computed analytically from saved logits at the first generation step.

580 **Spearman rank correlation.** Spearman’s ρ is the Pearson correlation coefficient between the token
 581 ranks of two distributions:

$$\rho = \text{Pearson}(\text{rank}(P_{\text{target}}), \text{rank}(P_{\text{base}})). \quad (5)$$

582 Because temperature scaling $P^{(\tau)} \propto P^{1/\tau}$ is a strictly monotone transform, the rank order of tokens
 583 is preserved exactly, so $\rho = 1$ for any $\tau > 0$ as a mathematical identity. Any value $\rho < 1$ therefore
 584 indicates rank reorganization beyond what temperature scaling can produce.

585 **Mass outside top- K .** Let $\text{TopK}(P_{\text{base}})$ be the set of K tokens with the highest probability under
 586 P_{base} . The probability mass that the candidate distribution places *outside* this set is

$$\text{MassOutside}_K = 1 - \sum_{t \in \text{TopK}(P_{\text{base}})} P_{\text{target}}(t). \quad (6)$$

587 This directly measures support expansion: how much probability the candidate assigns to tokens that
 588 are not in the baseline’s preferred set. Reported values use $K = 10$.

589 **Jensen–Shannon divergence.** The symmetrized, bounded version of Kullback–Leibler divergence:

$$\text{JS}(P, Q) = \frac{1}{2} \text{KL}(P \parallel M) + \frac{1}{2} \text{KL}(Q \parallel M), \quad M = \frac{1}{2}(P + Q). \quad (7)$$

590 We use base-2 logarithms so that $\text{JS} \in [0, 1]$. Tokens absent from the stored top-100 are assigned a
 591 sentinel logit (-10^9) before softmax, which is negligible for our setting since the top-5 covers more
 592 than 99.9% of the probability mass.

C Proofs for Energy-Theoretic Analysis

C.1 Deterministic Trapping

Lemma 1 (Descent lemma). *If E is L_E -smooth and $\eta \leq 1/L_E$, then $E(\boldsymbol{\sigma}^{\text{new}}) \leq E(\boldsymbol{\sigma}) - \frac{\eta}{2} \|\nabla E(\boldsymbol{\sigma})\|^2$.*

Proof. By L_E -smoothness, $E(\boldsymbol{\sigma} - \eta \nabla E) \leq E(\boldsymbol{\sigma}) - \eta \|\nabla E\|^2 + \frac{L_E \eta^2}{2} \|\nabla E\|^2 = E(\boldsymbol{\sigma}) - \eta(1 - \frac{L_E \eta}{2}) \|\nabla E\|^2$. Since $\eta \leq 1/L_E$, we have $L_E \eta/2 \leq 1/2$, giving $E(\boldsymbol{\sigma}^{\text{new}}) \leq E(\boldsymbol{\sigma}) - \frac{\eta}{2} \|\nabla E\|^2$. $\square$

Theorem 1 (Deterministic trapping). *Without perturbation, the forward pass converges to the local minimum $\boldsymbol{\sigma}^*$ in the basin of attraction of $\boldsymbol{\sigma}^{(0)}$, which may satisfy $E(\boldsymbol{\sigma}^*) > E(\boldsymbol{\sigma}_g^*)$.*

Proof. Lemma 1 guarantees monotone energy decrease at each layer: $E(\boldsymbol{\sigma}^{(l+1)}) \leq E(\boldsymbol{\sigma}^{(l)})$. Since E is bounded below (the quadratic term dominates), the sequence $\{E(\boldsymbol{\sigma}^{(l)})\}$ converges, implying $\|\nabla E(\boldsymbol{\sigma}^{(l)})\| \rightarrow 0$ and convergence to a critical point. Deterministic dynamics follow a unique trajectory from $\boldsymbol{\sigma}^{(0)}$ and cannot cross energy barriers to reach minima in other basins. $\square$

C.2 Properties of the RSP-Induced Perturbation

Decomposition of attention output. The attention update with n real keys and k random keys can be decomposed as

$$\mathbf{x}_i^{(l+1)} = (1 - w_{r,i}^{(l)}) \tilde{\mathbf{x}}_i^{(l+1)} + \boldsymbol{\eta}_i^{(l)}, \quad (8)$$

where

$$\tilde{\mathbf{x}}_i^{(l+1)} = \sum_{j=1}^n \tilde{\alpha}_{ij}^{(l)} \mathbf{v}_j^{(l)}, \quad \tilde{\alpha}_{ij}^{(l)} = \frac{\alpha_{ij}^{(l)}}{1 - w_{r,i}^{(l)}}, \quad (9)$$

and

$$\boldsymbol{\eta}_i^{(l)} = \sum_{j=1}^k \alpha_{i,n+j}^{(l)} \mathbf{v}_{r_j}^{(l)}. \quad (10)$$

We now analyze the properties of the perturbation term $\boldsymbol{\eta}_i^{(l)}$.

Conditional expectation. Conditioned on the attention weights, we have

$$\mathbb{E}[\boldsymbol{\eta}_i^{(l)} \mid \alpha^{(l)}] = \mu_E w_{r,i}^{(l)}, \quad (11)$$

where $w_{r,i}^{(l)} := \sum_{j=1}^k \alpha_{i,n+j}^{(l)}$.

Conditional covariance. Assuming the random vectors are independent with covariance $\sigma_E^2 \mathbf{I}$,

$$\text{Cov}(\boldsymbol{\eta}_i^{(l)} \mid \alpha^{(l)}) = \sigma_E^2 \sum_{j=1}^k (\alpha_{i,n+j}^{(l)})^2 \mathbf{I}. \quad (12)$$

Using $\sum_j \alpha_j^2 \leq (\sum_j \alpha_j)^2$, we obtain

$$\text{Cov}(\boldsymbol{\eta}_i^{(l)} \mid \alpha^{(l)}) \preceq \sigma_E^2 (w_{r,i}^{(l)})^2 \mathbf{I}. \quad (13)$$

Bias-to-variance ratio. The ratio between the mean and standard deviation is bounded by

$$\frac{\|\mathbb{E}[\boldsymbol{\eta}_i^{(l)} \mid \alpha^{(l)}]\|}{\sqrt{\text{Var}(\boldsymbol{\eta}_i^{(l)} \mid \alpha^{(l)})}} \leq \frac{|\mu_E|}{\sigma_E}. \quad (14)$$

Empirically, $|\mu_E|/\sigma_E \ll 1$, implying that the stochastic component dominates the bias.

Table 9: Embedding matrix statistics. $|\mu_E|/\sigma_E < 0.02$ across all models.

Model	μ_E	σ_E	$ \mu_E /\sigma_E$
LLaMA-3.1-8B-Instruct	0.00002	0.01057	0.002
Qwen2.5-Math-7B-Instruct	-0.00002	0.01647	0.001
Qwen2.5-Math-1.5B-Instruct	0.00019	0.01585	0.012
Qwen2.5-Math-7B	-0.00003	0.01842	0.001
Qwen2.5-Math-1.5B	0.00038	0.02990	0.013

Empirical verification. Table 9 reports the embedding matrix statistics for all models used in our experiments.

Implication. Therefore, the perturbation behaves as approximately zero-mean noise whose magnitude is controlled by $w_{r,i}^{(l)}$, justifying the Langevin-like interpretation in the main text.

C.3 Attention Decay Bound

Theorem 2 (Attention decay). *The total attention on random tokens satisfies*

$$w_{r,i}^{(l)} \leq \frac{k}{k + n \cdot \exp(\Delta_i^{(l)}/\sqrt{d})}, \quad (15)$$

where $\Delta_i^{(l)} := \min_{j \in [n]} \mathbf{q}_i^\top \mathbf{k}_j - \max_{j' \in [k]} \mathbf{q}_i^\top \mathbf{k}_{r,j'}$ is the real-vs-random logit gap.

Proof. Let $s_j = \mathbf{q}_i^\top \mathbf{k}_j / \sqrt{d}$ for real keys and $s_{r,j'} = \mathbf{q}_i^\top \mathbf{k}_{r,j'} / \sqrt{d}$ for random keys. By definition of $\Delta_i^{(l)}$, each real key satisfies $s_j \geq \Delta_i^{(l)}/\sqrt{d} + s_{r,\max}$ where $s_{r,\max} = \max_{j'} s_{r,j'}$. Therefore $\exp(s_j) \geq \exp(\Delta_i^{(l)}/\sqrt{d}) \cdot \exp(s_{r,\max})$ for every real key. Summing over n real keys:

$$\sum_{j=1}^n \exp(s_j) \geq n \cdot \exp(\Delta_i^{(l)}/\sqrt{d}) \cdot \exp(s_{r,\max}) \geq n \cdot \exp(\Delta_i^{(l)}/\sqrt{d}) \cdot \frac{1}{k} \sum_{j'=1}^k \exp(s_{r,j'}). \quad (16)$$

The total attention on random tokens is $w_{r,i}^{(l)} = \sum_{j'} \exp(s_{r,j'}) / (\sum_j \exp(s_j) + \sum_{j'} \exp(s_{r,j'}))$. Substituting the lower bound on the real-key sum and letting $R = \sum_{j'} \exp(s_{r,j'})$:

$$w_{r,i}^{(l)} = \frac{R}{\sum_j \exp(s_j) + R} \leq \frac{R}{n \cdot \exp(\Delta_i^{(l)}/\sqrt{d}) \cdot R/k + R} = \frac{k}{k + n \cdot \exp(\Delta_i^{(l)}/\sqrt{d})}. \quad (17)$$

□

C.4 Proofs for Output Distribution Shift

C.4.1 Autoregressive Formulation

In autoregressive decoding, the joint distribution over a sequence $y_{1:T}$ factorizes as

$$\Pr(y_{1:T} \mid x) = \prod_{t=1}^T p(y_t \mid x, y_{<t}),$$

where each step is governed by a prefix-conditioned distribution. Any perturbation introduced by RSP affects generation through a sequence of local changes to $p(y_t \mid x, y_{<t})$, rather than a single global distribution.

C.4.2 Local Linearization of Logits under RSP

Let h denote the random soft prompt. We model the perturbed logits as a function $z_i(h)$, assuming local smoothness in a neighborhood of $h = 0$:

$$z_i(h) = z_i(0) + \nabla_h z_i(0)^\top h + r_i(h),$$

where $r_i(h) = O(\|h\|^2)$. Defining $a_i := \nabla_h z_i(0)$, we obtain the local linear approximation $z_i(h) \approx z_i + a_i^\top h$. This approximation is valid for sufficiently small $\|h\|$ and is used to characterize the direction of perturbation rather than exact logit values.

642 C.4.3 Proof of Proposition 1 (Stochastic Ranking)

643 *If there exists a token pair (i, j) such that $a_i \neq a_j$, then $0 < \Pr_h(z_i(h) > z_j(h)) < 1$.*

644 *Proof.* Under the local linear approximation, $z_i(h) - z_j(h) = (z_i - z_j) + (a_i - a_j)^\top h$. Let
 645 $b_{ij} := a_i - a_j \neq 0$. If $h \sim \mathcal{N}(0, \sigma^2 I)$, then $b_{ij}^\top h \sim \mathcal{N}(0, \sigma^2 \|b_{ij}\|^2)$, a non-degenerate Gaussian.
 646 The perturbed logit difference $\Delta_{ij}(h) := (z_i - z_j) + b_{ij}^\top h$ therefore has non-zero probability mass
 647 on both sides of zero: $0 < \Pr_h(\Delta_{ij}(h) > 0) < 1$. $\square$

648 C.4.4 On Proposition 2 (Support Expansion)

649 Proposition 2 states that tokens outside the baseline top- K set can enter the top- K set under RSP
 650 with positive probability. Proposition 1 establishes that pairwise ranking between tokens becomes
 651 stochastic. Entering the top- K set requires surpassing multiple tokens simultaneously, i.e., conditions
 652 of the form $z_i(h) > z_j(h)$ for all $j \in J$ where J is a subset of higher-ranked tokens. Under the local
 653 linear model, these inequalities define a region in h -space. If this region has non-zero measure under
 654 the distribution of h , then top- K entry occurs with positive probability. We interpret Proposition 2 as a
 655 support-expansion statement: RSP enables tokens that are effectively inaccessible under deterministic
 656 ranking to become reachable under stochastic perturbations.

657 C.4.5 Trajectory-Level Implications

658 Since autoregressive generation composes conditional distributions across steps, perturbations at
 659 early steps can lead to distinct trajectories. Let $\mathcal{R}_{k,K}(x)$ denote the set of trajectories containing
 660 at least one token outside the baseline top- K set. If RSP increases the accessibility of such tokens
 661 at early steps, then the probability of sampling trajectories in $\mathcal{R}_{k,K}(x)$ increases under repeated
 662 sampling. This links local ranking perturbations to global trajectory diversity.

D Attention Mass Analysis

This appendix formalizes the per-token attention mass reported in Figure 3 and the exclusion criteria applied to the reasoning span and the layer axis. For each of the 500 MATH-500 problems, a fresh RSP $\mathbf{H} \in \mathbb{R}^{L \times d}$ with $L = 10$ is sampled independently, so the reported heatmaps average over both problem variability and RSP-vector variability.

D.1 Per-Token Attention Mass

For each problem, we measure attention on the concatenated sequence [prompt; $\mathbf{H}$; generation], where the generation is the trajectory produced by the same model under matching RSP injection. Let $\alpha_{i,j}^{(\ell,h)}$ denote the post-softmax attention weight at layer ℓ , head h , from query position i to key position j , satisfying $\sum_j \alpha_{i,j}^{(\ell,h)} = 1$ under the causal mask. Let Q and R denote the index sets of question and RSP tokens with sizes $|Q|$ and $|R| = L$, and let H be the number of heads. The per-token attention mass that query i directs to each group at layer ℓ is

$$\text{PTM}_Q[\ell, i] = \frac{1}{|Q|H} \sum_{h=1}^H \sum_{j \in Q} \alpha_{i,j}^{(\ell,h)}, \quad \text{PTM}_R[\ell, i] = \frac{1}{|R|H} \sum_{h=1}^H \sum_{j \in R} \alpha_{i,j}^{(\ell,h)}. \quad (18)$$

Normalization by $|Q|$ and $|R|$ controls for group size, so both quantities express how much attention a single question (resp. RSP) token receives on average under the same softmax denominator. Question tokens thereby serve as a size-matched baseline that absorbs sequence-length effects common to both groups.

D.2 Heatmap Aggregation

We partition the layer indices and the reasoning position indices each into five equal-size bins $\{B_L^{(b)}\}_{b=1}^5$ and $\{B_P^{(b)}\}_{b=1}^5$. For sample s and target group $\bullet \in \{Q, R\}$, the value in cell (b_L, b_P) is

$$M_\bullet^{(s)}[b_L, b_P] = \frac{1}{|B_L^{(b_L)}| \cdot |B_P^{(b_P)}|} \sum_{\ell \in B_L^{(b_L)}} \sum_{i \in B_P^{(b_P)}} \text{PTM}_\bullet[\ell, i]. \quad (19)$$

Figure 3 reports the sample average of Eq. (19) over the 500 valid samples.

D.3 Excluding the `\boxed{\dots}` Span

The reasoning position range terminates strictly before the final `\boxed{\dots}` span. Chain-of-thought outputs decompose into a *text* (content) component and a *pattern* (template) component that contribute independently to downstream performance [Madaan and Yazdanbakhsh, 2022]. The `\boxed{\dots}` wrapper is a canonical pattern: a short, stereotyped span whose role is to complete the answer format rather than to extend reasoning. Including it would therefore inject a template-driven attention signature unrelated to reasoning dynamics.

D.4 Excluding the Final Two Layers

The layer axis further excludes the last two transformer layers. Two well-documented properties of late layers, none specific to our setup, motivate this choice.

Output-projection specialization. Late-layer hidden states lie in an approximately linear relationship with vocabulary logits and therefore function as a progressive linear readout rather than as representation-transforming blocks [Belrose et al., 2023]. Consistently, late-layer feed-forward modules have been characterized as output-distribution shaping mechanisms [Geva et al., 2021]. Attention in these layers therefore reflects output formation rather than reasoning computation.

Register-token hypothesis. Uninformative patch tokens in Vision Transformers have been shown to be repurposed as *registers*: storage slots that absorb attention in deeper layers [Darcet et al., 2024]. Because RSPs share a similar profile as continuous tokens without pretrained semantic content, a

701 similar repurposing in late transformer layers cannot be ruled out. Excluding the final two layers
 702 removes this potential register-induced confound from the reasoning-phase signal.

703 The two lines of evidence above identify known late-layer phenomena — linear readout and potential
 704 register repurposing — whose attention signatures are orthogonal to reasoning-phase dynamics.
 705 Removing the final two layers preempts their contamination of the reasoning-phase signal, yielding a
 706 principled, literature-grounded exclusion rather than a model-specific tuning decision.

707 D.5 Additional Suffix Heatmaps

708 Figure 5 reports the same per-token RSP attention analysis under suffix injection for Qwen2.5-Math-
 709 1.5B-Instruct and LLaMA-3.1-8B-Instruct. For the late-layer reasons discussed above, the pattern
 710 does not generalize uniformly across every cell; nevertheless, the overall trend is consistent across
 711 both models: question-token attention varies broadly across layers, whereas RSP attention shows the
 712 layer-wise and sequence-wise decay predicted by Eq. (4).

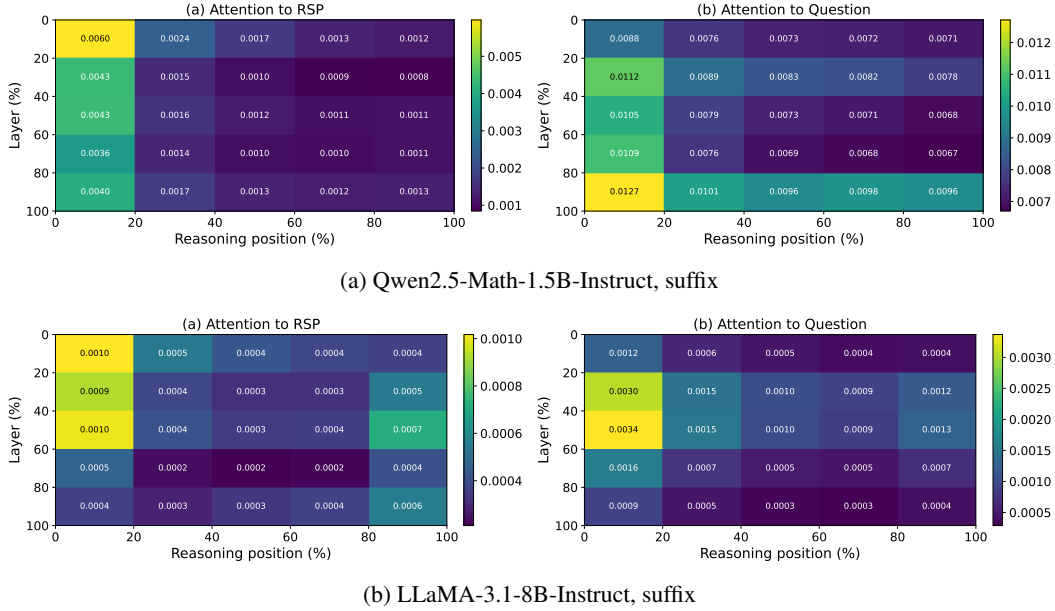


Figure 5: Per-token RSP attention mass under suffix injection for the remaining two models (500 MATH-500 samples each). Axes and preprocessing match Figure 3: x-axis is reasoning position binned into five quantiles, y-axis is layer depth binned into five quantiles (top = shallow), color encodes the 500-sample mean per-token attention assigned to RSP tokens, and tokens inside `\boxed{}` spans and the last two layers are excluded.

713 **E GRPO Formulation**

714 For a given prompt q , the behavior policy $\pi_{\theta_{\text{old}}}$ samples a group of G responses $\{o_i\}_{i=1}^G$ with
 715 corresponding rewards $\{R_i\}_{i=1}^G$. The advantage of each response is computed by normalizing the
 716 group-level rewards:

$$\hat{A}_{i,t} = \frac{R_i - \text{mean}(\{R_i\}_{i=1}^G)}{\text{std}(\{R_i\}_{i=1}^G)}. \quad (20)$$

717 The policy is updated via a clipped objective with a KL penalty:

$$\mathcal{J}_{\text{GRPO}}(\theta) = \mathbb{E} \left[\frac{1}{G} \sum_{i=1}^G \frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \left(\min \left(r_{i,t} \hat{A}_{i,t}, \text{clip}(r_{i,t}, 1-\varepsilon, 1+\varepsilon) \hat{A}_{i,t} \right) - \beta D_{\text{KL}}(\pi_{\theta} \parallel \pi_{\text{ref}}) \right) \right], \quad (21)$$

718 where $r_{i,t} = \pi_{\theta}(o_{i,t} \mid q, o_{i,<t}) / \pi_{\theta_{\text{old}}}(o_{i,t} \mid q, o_{i,<t})$ is the importance sampling ratio, ε is the clipping
 719 range, and β controls the strength of the KL penalty against the reference policy π_{ref} .

720 **F Broader Impacts**

721 This work provides empirical and theoretical analysis of how random embedding injection affects
722 the reasoning behavior of large language models. The contributions are primarily foundational,
723 with potential downstream implications for reinforcement learning-based post-training of reasoning
724 models such as GRPO.

725 **Positive impacts.** RSP-induced trajectory diversity could improve the sample efficiency of RL-
726 based LLM post-training by providing richer learning signals within group-relative advantage estima-
727 tion. This has the potential to reduce compute costs associated with alignment and reasoning-oriented
728 fine-tuning, making such methods more accessible to research groups with limited resources.

729 **Potential concerns.** Methods that expand the effective output support of LLMs may increase the
730 reachability of low-probability tokens, which includes both desirable alternative reasoning paths
731 and potentially unsafe or off-distribution outputs. Practitioners applying RSP in production systems
732 should combine it with existing safety filtering and alignment mechanisms rather than treating it as a
733 standalone diversity source.

734 **Limitations on impact scope.** Since this work focuses on analysis rather than deploying a new
735 model or dataset, the direct societal impact is limited. The broader implications described above are
736 contingent on future work integrating RSP into applied training pipelines.

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